

SATHURSHNA UTHAYAKUMAR

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🌐 [LinkedIn](#)

🐙 [GitHub](#)

SUMMARY

Motivated Computer Science and Engineering undergraduate with a strong interest in Software Engineering and full-stack development. Passionate about designing efficient, scalable, and user-friendly software solutions. Continuously learning modern technologies and eager to apply my knowledge through real-world projects and collaborative development.

EDUCATION

University of Moratuwa

BSc. Engineering (Hons) in Computer Science and Engineering

2024 – Present

Sri Lanka

- Current GPA: 3.28/4.0 (3rd year)

BT/Vincent Girls' High School

G.C.E. Advanced Level Examination – Physical Science Stream

2022 (2023)

Batticaloa, Sri Lanka

- Results: 2A B (Distinction)
- Z-Score: 2.1460
- Island Rank: 700

PROJECTS

AI-Powered Automated Inventory and Freshness Monitoring System for Small-Scale Retailers

Ongoing

🐙 [Repository](#)

Technologies: React Native, Expo, TypeScript, FastAPI (REST API), PostgreSQL (Supabase), Redis, Celery, Cloudflare R2, Docker, GitHub Actions

- Developing an AI-powered full-stack inventory and freshness monitoring system for small-scale retailers.
- Enables vendors to capture produce images and automatically classify freshness using AI.
- Supports real-time inventory management, low-stock and spoilage alerts, and streamlined retail operations.
- Provides a cross-platform mobile application for vendors and an administrative web dashboard.

Smart Blood Donation Coordination System

Ongoing

🐙 [Repository](#)

Technologies: React Native, Expo, JavaScript, Node.js, Express.js (REST API), PostgreSQL, JWT Authentication, bcrypt, Git, GitHub

- Developing a full-stack Blood Donation Coordination System with role-based authentication.
- Implementing RESTful APIs and PostgreSQL-based data management.
- Designed to efficiently coordinate donors, hospitals, and blood donation requests.

Industrial Predictive Maintenance System

March 2026 – June 2026

Technologies: React, FastAPI (REST API), PostgreSQL, InfluxDB, Apache Kafka, MLflow, Docker, Kubernetes, Google Cloud Platform (GKE), Keycloak, Kong API Gateway, Prometheus, GitHub Actions

- Developed a scalable IoT-based predictive maintenance system that collects industrial sensor data.
- Implemented edge processing and real-time data streaming for industrial monitoring.
- Developed machine failure prediction capabilities and visualized operational insights through a web dashboard.

Supply Chain Management System – Full-Stack Web Platform

September 2025 – October 2025

Technologies: React, TypeScript, Vite, Tailwind CSS, Radix UI, React Router, React Query, FastAPI, Python, MySQL, Unicorn, bcrypt

- Developed a full-stack logistics management application with dedicated customer and admin portals.
- Implemented secure authentication and database schema design.
- Developed RESTful APIs using FastAPI and Python to support application workflows.

TECHNICAL SKILLS

Programming: Python, Java, C, C++, JavaScript, TypeScript

Frameworks: React, React Native, FastAPI, Node.js, Django, Tailwind CSS, Vite

Databases: MySQL, PostgreSQL

Tools & Libraries: Git/GitHub, Prometheus

VOLUNTEERING EXPERIENCE

Hit the Ground

Organizing Committee – Design Committee

- Coordinated with participating companies and supported the organization and smooth execution of the CSE Career Fair.

LANGUAGE PROFICIENCY

English - Fluent · Tamil - Native · Sinhala - Basic

REFERENCES

PROF. DULANI MEEDENIYA

Department of Computer Science & Engineering

University of Moratuwa

Email: dulanim@cse.mrt.ac.lk

Current Semester Supervisor

DR. ADEESHA WIJAYASIRI

Department of Computer Science & Engineering

University of Moratuwa

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Industrial Training Coordinator